



PROJECT OVERVIEW

Project Overview

Architecture, controls, security, deployment, and project status.

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Move your hand. Play the game.

VirtualGlove turns hand movement and gestures into responsive RetroArch controls using an ordinary USB camera and a **VirtualGlove Controller** built on the Arduino UNO Q. Wear a plain glove or use your bare hand-there are no sensors, wires, or electronics to add to it.

Move to steer. Curl fingers for buttons. Roll, push, pull, grab, throw, and punch. VirtualGlove recognises the pose, sends authenticated controller input across your local network, and lets RetroArch see controller input or a native Power Glove controller.

Current stable release: v0.5.1. It includes smoother Batocera Wi-Fi input delivery, the revised Controller Dock V2.1, and verified release-package checksums. Use the same installation commands for a fresh setup or an upgrade.

VirtualGlove supports RetroPie, Recalbox 10.x, Batocera 38+, and LaunchBox on 64-bit Windows. Recalbox and Batocera use **Controller Router** to combine configured physical controllers and VirtualGlove as enabled merged Players 1-4. RetroPie can opt into the same Router while retaining its separate-gamepad default. LaunchBox keeps physical XInput beside VirtualGlove's managed RetroArch controller. Every platform keeps the physical controller usable and preserves unrelated controller settings.

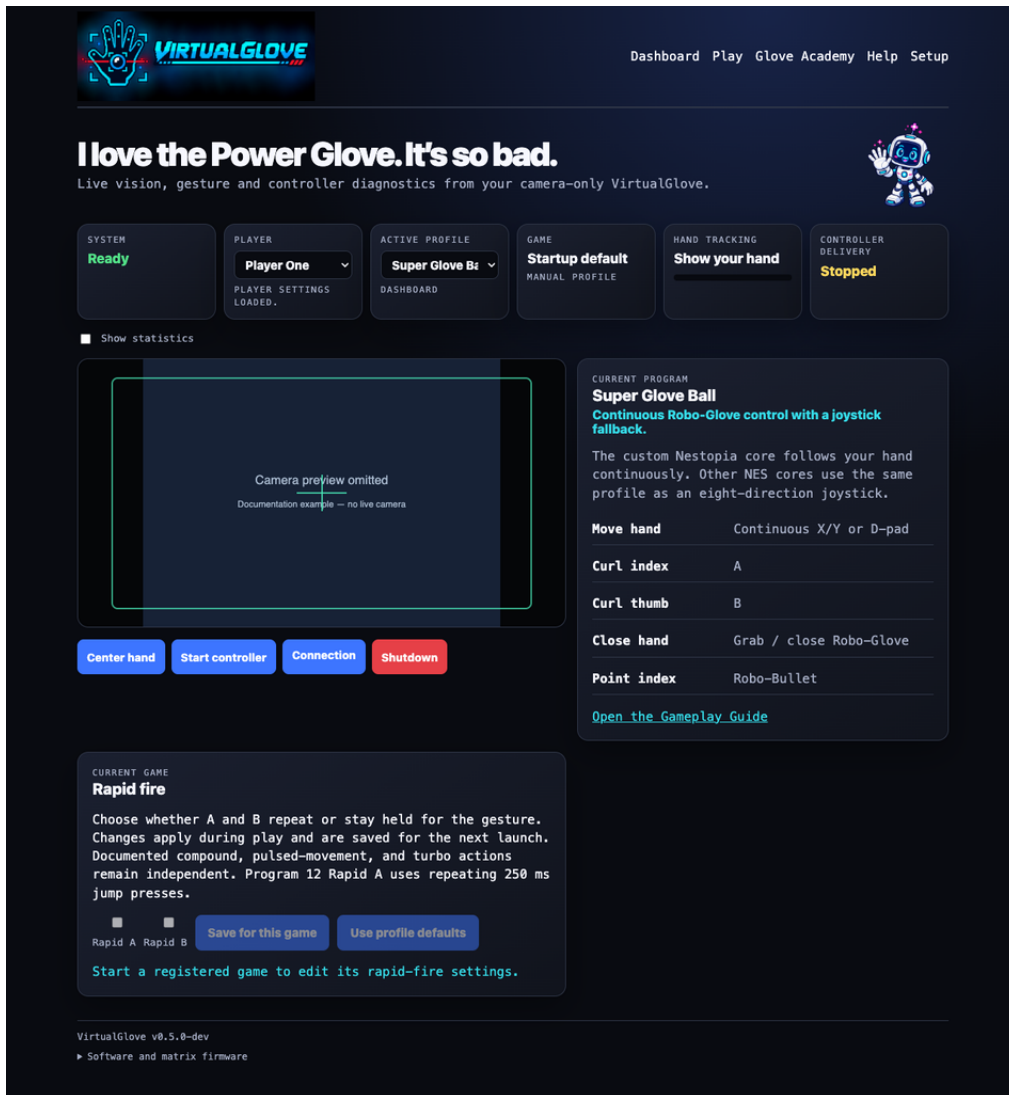
On routed RetroPie, Recalbox, and Batocera, assigned physical controllers keep their EmulationStation button mappings and player assignments in every RetroArch/Libretro system. VirtualGlove gestures remain intentionally narrower: ordinary joystick gestures control supported NES cores, while native Super Glove Ball uses its separate guarded input path.

When Controller Router is enabled, a game launch always begins neutral. Physical controllers are available immediately; VirtualGlove joins only after the hand has returned to neutral once. This prevents a gesture observed in EmulationStation from becoming the game's first input.

Ordinary NES games use FCEUmm. Super Glove Ball can also use the separately named Nestopia (VirtualGlove) core for native movement and glove actions without replacing stock Nestopia. The current installers provide a guarded upgrade from v0.4.2 while preserving players, calibration, tuning, pairing, device settings, game registrations, ROMs, saves, and controller assignments.

Why VirtualGlove?

- **Camera-only play:** use a standard UVC camera rather than modifying a glove.
- **Two styles of NES control:** ordinary joystick output through FCEUmm and continuous native movement for Super Glove Ball through [lr-nestopia-powerglove](#).
- **Original programs and game mappings:** Programs 1-14 and A-I, including Mattel's indexed games and documented rapid-fire exceptions.
- **Fast, direct tracking:** MediaPipe Hands processes the newest camera frame and sends the latest valid hand coordinate without a settling tail.
- **Family-friendly learning:** Pixel Pal guides players through 16 Glove Academy lessons without sending accidental input to a game.
- **Personal setup:** each player can save a hand centre, movement reach, joystick centre box, gesture sensitivity, and Academy progress.
- **Game-aware and safe:** registered games select their profile automatically; stale, lost, or unauthenticated input returns to neutral.
- **Local by design:** video stays on the Controller. Pairing and controller traffic are authenticated between your own devices.



VirtualGlove Dashboard with camera and controller status

What you need

- An Arduino UNO Q provisioned through Arduino App Lab
- A supported console: RetroPie, Recalbox 10.x, Batocera 38+, or LaunchBox with 64-bit Windows and 64-bit RetroArch
- A UVC-compatible USB camera and powered USB hub
- A physical controller for RetroArch setup and recovery
- Both devices on the same trusted local network with internet access during installation
- Your own legally obtained games-VirtualGlove includes no ROMs or BIOS files

New to the hardware? Start with [Build your own](#) for the parts, expected cost, and difficulty.

Install VirtualGlove

These steps install the latest stable release. Install the **same version on both devices** and close any running RetroArch game first. The scripts verify their downloads, ask for administrator access when needed, and preserve existing pairing and player settings during an update.

VirtualGlove **v0.5.1** is the current stable release. The unversioned commands below select it through GitHub's latest stable release. To reproduce one exact installation later, use the pinned-version procedure in the [technical installation reference](#).

1. Prepare the Controller

Finish the UNO Q's App Lab setup, connect it to your network, and attach the camera through the powered hub. The camera may also be connected after installation.

2. Install the Controller software

Open a terminal on the UNO Q and run:

```
SH
cd /home/arduino
curl -fLO \
  https://github.com/mathan416/VirtualGlove/releases/latest/download/install-uno-q.sh \
  && bash install-uno-q.sh
```

On a first installation, the installer suggests **virtualglove** as the Controller name, making its usual address **virtualglove.local**. Press Enter to accept it or type a different family-friendly name. Updates preserve the existing name.

3. Check the Controller

Open the Dashboard address printed by the installer—normally **http://virtualglove.local:8088/dashboard**. The installer configures automatic startup, the matrix display, guarded camera recovery, and its required host helpers.

4. Install or upgrade the console software

Choose only the section for your game system. Close every running RetroArch game first. These are both installation and upgrade instructions: rerun the same command or LaunchBox installer whenever a new VirtualGlove version is available. Existing ROMs, saves, pairing, game registrations, controller assignments, and unrelated emulator settings are preserved.

RetroPie

Connect as the normal RetroPie user, not through a root shell, and run:

```
SH
cd "$HOME"
curl -fLO \
  https://github.com/mathan416/VirtualGlove/releases/latest/download/install-retropie.sh \
  && bash install-retropie.sh
```

On a first installation, enter the Controller name or address—normally **virtualglove.local**. The installer can add missing emulator support, the optional native Super Glove Ball core, and the optional calibration test. A normal RetroPie installation keeps VirtualGlove as a separate gamepad; Controller Router remains an explicit opt-in for multi-controller

cabinets.

Recalbox 10.x

Connect over SSH as [root](#). Do not add [sudo](#). Run:

```
SH
cd /recalbox/share/system
curl -fLO \
  https://github.com/mathan416/VirtualGlove/releases/latest/download/install-recalbox.sh \
  && bash install-recalbox.sh
```

On a first installation, enter the Controller name or address and select the physical Player 1 controller when more than one configured controller is available. Recalbox creates **VirtualGlove Merged Player 1** initially; after pairing, Setup can assign configured controllers and VirtualGlove across merged Players 1-4.

Batocera 38 and newer

Connect over SSH as [root](#). Do not add [sudo](#). Run:

```
SH
cd /userdata/system
curl -fLO \
  https://github.com/mathan416/VirtualGlove/releases/latest/download/install-batocera.sh \
  && bash install-batocera.sh
```

On a first installation, enter the Controller name or address and select the physical Player 1 controller when needed. Batocera creates **VirtualGlove Merged Player 1** initially and uses the same Setup-managed Player 1-4 routing model as Recalbox.

LaunchBox on Windows x86-64

Install 64-bit Python and 64-bit RetroArch with FCEUmm first. Confirm the physical XInput controller works, then close LaunchBox, Big Box, and RetroArch. Download [VirtualGlove-LaunchBox.zip](#) from the latest release and extract it. Open PowerShell as Administrator using the same Windows account that runs LaunchBox, enter the extracted [VirtualGlove](#) folder, and run:

```
POWERSHELL
Set-Location "$env:USERPROFILE\Downloads\VirtualGlove"
powershell -ExecutionPolicy Bypass -File .\launchbox\install-launchbox.ps1 `
  -LaunchBoxRoot "C:\LaunchBox" -RetroArchRoot "C:\RetroArch" `
  -ControllerHost "virtualglove.local"
```

Change the extraction, LaunchBox, RetroArch, and Controller locations to match your computer. LaunchBox gains a **VirtualGlove RetroArch** NES emulator. Ordinary games use its managed loopback RetroPad while physical XInput and the real keyboard remain available; exact registered Super Glove Ball filenames can use Nestopia (VirtualGlove).

Check the console installation

The final Linux report should show no [FAIL](#) entries. [ACTION](#) is normal for pairing or physical gameplay checks that still require you. After pairing:

1. Test an ordinary registered NES game through FCEUmm with VirtualGlove and the physical controller.
2. On Recalbox, Batocera, or routed RetroPie, test one non-NES Libretro game with the physical controller.

3. Test native Super Glove Ball separately when installed.
4. Confirm the physical hotkey can still exit the game.

The complete [Installation Guide](#) explains every installer question, checkpoint, later-added ROM, update, and troubleshooting path.

5. Pair the devices

Open the secure Setup address printed by the installer—normally <https://virtualglove.local:8443/setup>—open **Connection and startup**, choose RetroPie, Recalbox, Batocera, or LaunchBox, save the console hostname or IP address, and continue into **Pair this Controller**. Pairing stays unavailable until the platform and address are both saved. The guided one-time-code method is recommended and shows only the command for the selected platform.

After confirming the browser certificate against the physical Matrix ID, the optional **Trust this Controller** step removes future privacy warnings on that phone or computer.

6. Set up a player

Choose a player, position the camera, and use **Centre hand**. Open **Glove Academy** to learn the gestures and adjust movement reach or sensitivity only if needed.

7. Play

Select **Start controller**, launch a registered game, and confirm the expected profile. FCEUmm uses joystick mode; Super Glove Ball can also use the optional native core selected per ROM on RetroPie, Recalbox, Batocera, or LaunchBox.

ROMs added later do not require reinstalling VirtualGlove. Refresh the console frontend's game list, then open **Setup -> Games** and register the exact ROM filename if it is not already one of the supplied aliases. RetroPie uses its per-ROM launch choice for native Super Glove Ball; Recalbox and Batocera apply their exact-ROM native choice at VirtualGlove service startup; LaunchBox's VirtualGlove RetroArch wrapper chooses the core on every launch. The complete installation guide lists the platform-specific refresh and restart steps.

The complete [Installation Guide](#) has first-install checkpoints, illustrated pairing, camera advice, native-core setup, updates, backups, and troubleshooting. Use it as the authoritative setup reference.

What can you play?

VirtualGlove includes the original Programs 1-14, nine reusable cartridge Programs A-I, plus dedicated mappings for Bad Street Brawler and Super Glove Ball. The same recognition settings follow the player across games; profiles change only what the recognised movements and gestures send to the console.

Path	What it provides
FCEUmm	The selected Programs 1-14 or A-I mapping, including positional movement, mapped A/B actions, and documented compound gestures.
Super Glove Ball with FCEUmm	A complete joystick-mode fallback that can always be selected for testing or play.
Super Glove Ball with Nestopia (VirtualGlove)	Continuous native X/Y and Z, Start, grab/catch, release/throw, Robo-Bullet fire, and Power Punch. The isolated core is supported on RetroPie, Recalbox, Batocera, and LaunchBox; FCEUmm remains the fallback.

Rapid A/B defaults on only where an individual program description explicitly identifies a pulsed button: Program 7 A, Program B A, Program H A/B, and Bad Street Brawler B. Every other profile defaults off. The registered Blaster Master, Double Dribble, Racket Attack, Ice Hockey, and Alpha Mission entries retain Mattel's documented off instructions. Programs 13 and 14 also support deliberate physical-controller and gestures-off play.

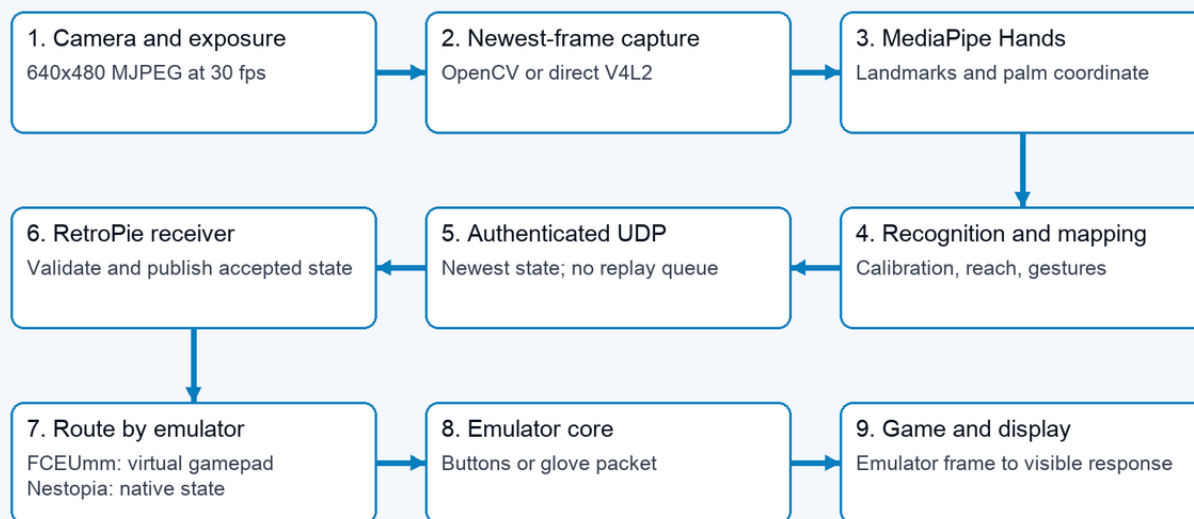
Dashboard Rapid A/B switches control button repetition only; profile-owned fast turns, pulsed movement, turbo movement, and compound actions retain their own timing. Saved per-game overrides survive upgrades. Choose **Use profile defaults** while that game is running to remove its overrides and adopt the corrected defaults.

The [Gameplay Guide](#) shows every numeric Program gesture, compound action, release rule, indexed game, exception, objective, and practice challenge. The [VirtualGlove Input Modes](#) explains why the joystick and native emulator paths feel different and how physical Player 1 controls join each platform.

The numeric Program section includes gesture-by-gesture controls, compound action timing, the complete official game index, and the five automatic rapid-fire exceptions. Program 13 is the mixed physical-controller option; Program 14 deliberately turns camera gestures off for the active game.

How it works

08 / Camera to game: one front end, two delivery paths



The browser preview and statistics observe the worker; neither belongs in the controller delivery path.

End-to-end VirtualGlove flow from camera to game

1. The camera delivers its newest frame to the VirtualGlove Controller.
2. MediaPipe Hands finds the palm, wrist, and finger landmarks.
3. Shared recognition turns those landmarks into position, fingers, rolls, depth motion, and menu poses.
4. The Controller sends the newest authenticated state to the selected console.
5. The console publishes either Player 1 input or native glove state for the selected emulator core.

Pairing is tied to a private shared key rather than one permanent IP address. If DHCP changes an address or [.local](#) resolution temporarily fails, the paired devices can rediscover one another on the local network without broadcasting controller states or requiring a new pairing.

Meet Pixel Pal

Pixel Pal helps players learn, personalise, test, and troubleshoot without turning setup into an engineering exercise.

- **Glove Academy** teaches all 16 movements and gestures while game output is paused.
- **Tune gestures** records guided examples and previews a conservative adjustment before saving it.
- **Find the best camera settings** compares only choices supported by the attached camera and changes nothing until the player accepts a recommendation.
- **Rock Paper Scissors** provides a camera-controlled practice game that does not require a connected console.

Controller pages

Page	Purpose
Dashboard · /dashboard	See live controls and set per-game A/B rapid fire.
Play · /play	Challenge Pixel Pal to Rock Paper Scissors.
Glove Academy · /learn	Learn gestures, set movement reach, and personalise recognition safely.
Setup · /setup	Manage players, camera choices, console pairing, games, backups, and display preferences.
Help · /help	Read the complete manuals and printable PDFs directly on the Controller.

Documentation

Start here

You want to...	Read...
Install, pair, and play your first game	Installation Guide
Learn gestures, Programs, and game controls	Gameplay Guide
Understand joystick, merged-controller, and native modes	VirtualGlove Input Modes
Choose or troubleshoot a camera	Camera Guide
Assemble a printed case quickly	Enclosure Assembly Quick Reference
Print or customise a case	Controller Enclosure Guide
Recognise matrix animations and messages	Matrix Display Guide
Find a quick command or status reminder	Quick Reference
Solve a problem by symptom	Troubleshooting

Go deeper

You want to...	Read...
See the current components and data flow	Architecture
Look up every setting and command	Configuration Reference
Follow the engineering process, experiments, and movement validation	Engineering Journey
Repeat camera, latency, trace, or native research	Engineering Toolkit
Review ROM-level native and joystick evidence	Power Glove Game ROM Input Audit
Review security and pairing boundaries	Security Policy
Check dependencies and third-party terms	Third-party Notices
Contribute code or documentation	Contributing Guide

Printable editions of all maintained guides are available in [output/pdf/](#). The Controller serves the same documentation from its local Help page.

The public website is maintained with the application under [website/](#). Its dependency-free builder reads [config/release.json](#), renders five static pages, validates release-sensitive links and commands, and creates [output/website/VirtualGlove-Website.zip](#) for manual upload.

Project status

VirtualGlove 0.5.1 keeps the proven CPU MediaPipe Hands path, Programs 1-14, live dead-zone visualization, and source-accurate rapid-fire behaviour. It extends the authenticated console integration to Recalbox, Batocera, and LaunchBox while retaining RetroPie. This release also smooths Batocera Wi-Fi input delivery and includes the revised Dock V2.1.

The 0.5.x release line completed physical controller and VirtualGlove acceptance on RetroPie, Recalbox 10.1.1, Batocera 43.1, and LaunchBox. This includes ordinary NES play, native Super Glove Ball, physical-controller coexistence, hotkeys, reboot persistence, and Controller Router remapping. Recalbox's all-Libretro physical path has also been exercised with Game Boy, ColecoVision, and Game Gear games.

Recalbox and Batocera use Controller Router to publish enabled merged Players 1-4 from configured physical sources and at most one VirtualGlove. Standard RetroPie keeps its separate gamepad until Router is explicitly enabled; the project arcade cabinet now uses Router for its two I-PAC interfaces, 8BitDo controllers, and VirtualGlove. LaunchBox uses a local RetroPad while leaving XInput and real keyboard controls available. Every platform supports registered FCEUmm games and the separately named Nestopia (VirtualGlove) path for native Super Glove Ball. The installers update managed files while preserving pairing, players, calibration, tuning, Academy progress, games, ROMs, and saves.

Use Setup's **Download system report** when asking for help. It records useful software, camera, controller, and connection health without including video, pairing keys, player calibration, ROM names, or network addresses.

Contributing and licensing

Issues, careful test reports, documentation improvements, and code contributions are welcome. Please read the [Contributing Guide](#) before opening a change. Release history is kept in the [Changelog](#).

VirtualGlove is maintained by **Iain Bennett** and is licensed under the [MIT License](#). The modified Nestopia core is GPLv2 software and remains separate from the MIT application. Nintendo, NES, Power Glove, and the named games belong to their respective owners. See [Third-party Notices](#) for dependency, model, asset, and native-core licensing details.